ComicCrafter AI Documentation

Overview

ComicCrafter AI is a generative AI-based comic generator that runs locally on edge devices. It transforms user prompts into comic-style stories by leveraging:

- Mistral LLM for story generation
- Stable Diffusion for image creation
- Streamlit for the web interface

Prerequisites

- Python 3.10
- NVIDIA GPU (tested with RTX 3060)
- 8GB+ RAM
- Git

Installation

Clone the repository:

```
git clone https://github.com/codingbot221/ComicCrafterAI.git
cd ComicCrafterAI
```

Create a virtual environment:

Windows:

```
python -m venv .venv
.venv\Scripts\activate
```

Linux/Mac:

```
python3 -m venv .venv
source .venv/bin/activate
```

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Install dependencies:

pip install -r requirements.txt

Usage

Run the application:

streamlit run app/main.py

Enter a prompt:

"A girl discovers a magical portal in her school library."

Generate comic:

The application will process the prompt and generate a comic-style story with corresponding images.

Technologies Used

Component	Tool Used and Purpose
Story Generation	Mistral LLM - Convert prompts into structured plots
Image Generation	Stable Diffusion - Generate comic-style images
UI/UX	Streamlit - Interactive web interface
Local Execution	Edge device - No cloud dependency, privacy focused

Project Structure

```
ComicCrafterAI/
├── app/
│   └── main.py          # Streamlit UI and logic
├── models/              # Model loading/configuration scripts
├── static/              # Static assets: images, fonts, styles
├── .venv/               # Virtual environment
├── requirements.txt     # Python dependencies
└── README.md            # Project overview and instructions
```

Requirements

- torch
- diffusers
- transformers
- PIL (Pillow)
- matplotlib
- fpdf
- streamlit